

Response to the reviewers

Manuscript CAOR-D-26-01885, “Optimal-stopping recourse for vehicle routing with simultaneous pickup, delivery, and stochastic demands”

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Dear Editor, dear Reviewers,

We thank the three reviewers for careful and constructive reports. They asked us to clarify the stochastic assumptions and the depot-return model, to test the method under dependent and non-exchangeable demand and under a shared standby pool, to compare against stronger threshold policies, to report comparable data and computing budgets, to strengthen the statistical analysis, and to state the contribution and the scope of the conclusions more carefully. We have addressed each of these points. Several of the new experiments did not come out entirely in our favour, and we report those results in the manuscript as they are: an equal-data three-action dynamic program ties BATON on the deterministic plans and beats it under very strong demand correlation, and on two city variants BATON is only tied with its strongest competitor. Every comment is answered below; the reviewers’ text is set in italics, and section, table and proposition numbers refer to the revised manuscript. A version of the manuscript with all changes marked against the submission is provided separately.

Summary of the main changes

1. **Assumptions and theory** (Section 3.2, Section 3.3, Section 3.4). Assumption 1 now requires only that the load be a stochastically monotone Markov chain, which covers independent increments and a Gaussian day-factor model; Assumption 2 now requires only that the emergency price be non-increasing along the route—no ordering between handoff and emergency prices is needed. Proposition 4 (monotone continuation value) is proved under these weaker assumptions and for the three-action recursion. A new Proposition 3 quantifies the cost of over-triggering (Reviewer 2, comment 4).
2. **New experiments** (Section 4.5–Section 4.7): demand correlation $\rho \in \{0, 0.3, 0.6, 0.9\}$ and a day-level demand factor, with a direct test of monotonicity and of its power; an exact dynamic program under independent demand; promotion days and day-type-specific fits; the bias of the fresh-start value and the exact post-return state; a capacitated standby pool shared by the routes of a plan or of a metropolitan group of plans, with a sweep of holding costs and a shadow price; training-data budgets from 100 to 20,000 days; single-core timing of every policy; deliver-only customers; city plans re-planned under the SAA gate; a configuration in which standby is dearer than an emergency; and a fixed fee on every depot return, with the fee at which the return lever stops paying (29 currency units).
3. **New competitors and references**: a position-dependent threshold (one tuned level per stop), a cost-scaled rollout (a cost-dependent threshold with a tuned price multiplier), a *two-lever* rule that gives a tuned threshold the depot return as well, and an equal-data dynamic program for the *three-action* problem (DP_N^3), so that BATON is compared with policies that have its own action set. As references we added a high-data plug-in program for the three-action problem, the three-action clairvoyant bound, and an *exact* dynamic program, solved

by numerical convolution, for the case of independent demand (Table 8).

4. **Statistics:** the instance within a gate is now the experimental unit (the routes of a plan share their days), with paired mean differences, 95% bootstrap confidence intervals, win counts, effect sizes and Holm-adjusted tests per gate (Table 3), paired intervals for the key contrast of every robustness table, tail-risk measures (Table 4) and route-level results as supplementary data.
5. **Positioning and scope:** the introduction separates the modelling contribution, the structural results and the adaptation of established regression-based stopping methods; the conclusion states the scope of the managerial conclusions and the boundaries of the action set.
6. **Length:** five supplementary tables, one figure and the proof of Proposition 1 have moved to an appendix.
7. **Machine-checked proofs:** all four propositions have been verified formally in the Lean 4 proof assistant with the Mathlib library. The formalisation compiles without errors or unproved steps and relies only on Lean’s standard axioms. It is released with the replication code (directory `papers/baton/BatonProofs`), and its correspondence to the text is documented there. The formalisation abstracts the transition law to a kernel with the monotonicity property, and proves Propositions 2 and 3 on finite trees of histories; the manuscript now lists these abstractions (Section 3, first paragraph).

In the course of the revision we also found and corrected several errors of our own; they are listed at the end of this letter. The most important is that the proof of the cost bound in the original Proposition 1 compared the optimal policy with the clairvoyant bound in the wrong direction. The corrected Proposition 1 gives two-sided bounds that are now proved.

Reviewer 1

Stochastic assumptions and state sufficiency. The manuscript should clarify the independence assumptions required for the theoretical results. In particular, the authors should justify under which conditions the route position and accumulated load constitute a sufficient state. An experiment with correlated demands or route-level demand factors would strengthen the analysis.

Response. We agree that the original Assumption 1 (independent increments) was stronger than needed, and that its relation to the correlated experiments was not explained. The revised Assumption 1 asks that the running load (W_k) be a Markov chain whose transition kernels are stochastically monotone. Under this condition (k, W_k) is a sufficient state for the handoff problem and all results go through; for the three-action problem it is sufficient under independent increments, and the manuscript now says after Eq. (14) that the recursion is misspecified under factor dependence, which the state-conditional variant BATON-CF corrects. The new text identifies two cases of practical interest: (i) independent, not necessarily identically distributed increments, and (ii) positively dependent demand driven by a common day-level factor, $g_j = \mu_j + \beta Z + \varepsilon_j$ with Gaussian Z and noise of common variance. In case (ii) the likelihood depends on Z only through W_k , so W_k is sufficient for the factor, its posterior is increasing in W_k , and the chain is Markov with monotone kernels. Outside these cases—including our benchmark protocol, where pickups and deliveries load on separate factors with customer-specific scales— W_k is not exactly sufficient, and we now say so and describe the policy as the regression-based approximation within the class of policies that observe (k, W_k) .

We then measured what this approximation costs. Table 7 re-fits and re-tests every policy on the

Dethloff DET and SAA plans under copula correlations $\rho = 0, 0.3, 0.6, 0.9$ and under a route-level multiplicative day factor. Dependence turns out to help rather than hurt: every implementable policy saves more as ρ grows, because correlated demand creates heavy days whose breaches can be foreseen from the running load. A BATON variant has the highest saving among the implementable policies under every law. BATON itself leads the strongest competitor with a paired interval above zero in six of the ten rows and is tied in three; in one row it loses: at $\rho = 0.9$ on DET plans the equal-data three-action dynamic program is ahead by -1.1 points ([-1.6, -0.7]). That is where the dependence bias of the fresh-start value is strongest, and BATON-CF reverses it (37.6% against 36.5%). On the DET plans BATON attains 100%, 94% and 93% of the high-data three-action saving at $\rho = 0, 0.6$ and 0.9 . Under independence we can also compute the exact optimum by numerical convolution (Table 8): on DET plans BATON attains 94% of the exact three-action saving. We also tested the shape directly: the unconstrained binned estimate of $\mathbb{E}[\text{future cost} \mid W_k]$ on fifty thousand paths per route decreases significantly between adjacent bins in 0.2% of pairs under independence, where monotonicity is a theorem and this rate measures the false-positive level of the test, in 0.1%, 0.1% and 0.0% of pairs at $\rho = 0.3, 0.6$ and 0.9 , and in 0.6% under the day factor. The test found no significant violations beyond its false-positive level, but its power is limited: a dip of 10% (25%) of the mean continuation cost injected into one bin is detected in only 64% (89%) of the tests. The evidence is therefore consistent with a monotone continuation value, without proving it, and the manuscript says so.

The main place where dependence matters is a finding we did not anticipate. It concerns the depot return: under dependence the value of the post-return state depends on the state that triggered the return, and an unconditional fresh-start value misprices it on exactly the days on which it is used. This is discussed under Reviewer 2, comment 2, and a state-conditional variant (BATON-CF) removes the effect.

Changes. Revised Assumption 1 and the paragraph after it (Section 3.2); proof of Proposition 4 (Section 3.4); new Section 4.5 with Table 7.

Shared standby capacity. The policy appears to be applied independently to each route, although standby vehicles are described as a pooled resource. If standby capacity is limited, decisions across routes are coupled. The authors should clarify whether capacity is unlimited, allocated in advance, or represented through an exogenous shadow price.

Response. In the original model standby capacity is pay-per-use: the pool is shared across a metropolitan operation and large relative to one plan, and each handoff is billed one standby vehicle-day (F_{sb}), so no plan exhausts it. We now state this explicitly and report the pool size each policy would need (the 95th percentile of daily handoffs per plan: for BATON a median of 3.5 and a maximum of 7 vehicles on the Dethloff DET plans, and a median of 0 on the SAA plans).

We agree that the capacitated case is important and have added it as a new experiment (Section 4.6, Table 10). A pool of S standby vehicles is reserved for the day, so a handoff then costs only its marginal price, and the pool serves either the routes of one plan or, in a metropolitan setting, the DET plans of all ten instances of a Dethloff class (about 66 routes). Requests are served first come, first served in the order in which they arise along the routes, and a route whose request is refused decides at the refusal stop between continuing and a depot return. We compare a naive policy that ignores the cap with a policy that decides each route with a shadow price λ on the pool, chosen on the training days, and with the two fixed prices $\lambda = F_{sb}$ and $\lambda = \infty$; we report the break-even holding cost of a reserved vehicle and the cost-minimising pool size at holding costs of 20 and 5 per vehicle-day. Three results follow. First, a reserved vehicle serves at most one route a day, so at a holding cost equal to the standby day rate a reserved pool cannot

beat pay-per-use, and in our data it never does; below that rate, reservation pays where handoffs are frequent. At a holding cost of 5 the optimal reserved pool beats pay-per-use on 40/40 Dethloff DET plans (119.0 against 135.0 per plan-day) and on 4/4 metropolitan pools, but not on the slack SAA plans or the city plans, which need few handoffs. Second, given a cap the routes are coupled, and deciding each route with a shadow price beats ignoring the cap wherever the cap binds (by 4.2% on Dethloff DET plans with one reserved vehicle, with a paired interval above zero, and by up to 9.4% on the metropolitan pools). Third, the shadow price chosen on the training days is high for small pools and falls as the pool grows, and neither fixed price matches it. The shadow price keeps the routes decoupled, so the policy is still fitted one route at a time, and the depot return is the lever that makes a small or empty pool tolerable. The three options the reviewer lists are therefore all covered: unlimited pooled capacity (the default model), capacity allocated in advance (the reserved pool, with its size and break-even cost), and an exogenous shadow price (the λ that decouples the routes).

Changes. New paragraph “Standby versus emergency vehicles” in Section 3.1; new Section 4.6 with Table 10; the conclusion lists shared resources among the scope limits.

Depot-return transition. The state transition after returning to the depot should be described more precisely. The manuscript should explain the vehicle load after unloading and reloading, the resulting residual capacity, whether multiple returns are allowed, and how this action affects the continuation-value recursion.

Response. Thank you; the original description was too brief and, in one respect, imprecise. The revised Section 3.5 now opens with a paragraph on the transition. After stop k the vehicle carries $L_0 + W_k$. A return unloads all pickups collected so far and leaves on board exactly the undelivered cargo $L_0 - D_{\leq k}$. The policy records the post-return state as $W = 0$ against the unchanged slack B , which is a conservative convention: the exact post-return state is $x_k = -D_{\leq k}$, so the convention understates the residual capacity by the volume already delivered. We adopt it because it fixes the post-return state and keeps the fresh-start value a single number per stop. Returns may be repeated: after a return the vehicle re-enters the recursion in state $(k, 0)$, so further returns and a later handoff remain available, and the recursion acquires the single extra term $R_k + C_k(0)$ in (14). To quantify the convention, we implemented the exact reset, with a fresh-start value $F_k(x)$ that depends on the observed delivered volume (Table A.3, column “exact reset”). Without deployment selection, pricing the exact residual capacity raises the three-action saving from 52.2% to 57.9% on Salhi–Nagy and from 51.4% to 54.6% on Dethloff SAA plans. The convention used in the main tables therefore understates the value of the depot return, and the reported results for the three-action policy are conservative. We kept the conservative convention in the main tables to preserve comparability with the submitted version and with the published restocking competitor, which uses the same reset. We state the exact reset as the recommended specification for deployment.

Changes. New paragraph “The depot-return transition” in Section 3.5; exact-reset results in Section 4.5 and Table A.3.

Threshold-policy comparison. The theoretical criticism applies mainly to fixed global thresholds. More flexible policies, such as position-dependent or cost-dependent thresholds, may remain competitive. Including such a benchmark would help isolate the value of estimating the full continuation function.

Response. This is a good point, and we have added both, together with a third benchmark that the question suggests. The *position-dependent threshold* (thr.- k) has one load threshold per

stop, tuned by coordinate descent on realised training cost. Because C_k is monotone, the optimal stopping region at each stop is an upper set $\{w > w_k^*\}$, so this class can represent the optimal handoff boundary exactly, and it isolates precisely what the reviewer asks: the value of *estimating* the boundary by backward induction rather than searching for it. The coordinate descent starts from the best of the reactive policy, the per-stop cuts of the tuned global threshold and the per-stop boundaries of BATON-ho, so it is at least as good as each of them on the training days. The *cost-scaled rollout* (roll- θ) hands off when the myopic cost $C_k^0(W_k)$ exceeds θH_k with a tuned θ ; this is a cost-dependent threshold that uses the per-stop prices. Finally, the *two-lever* rule gives a tuned threshold the depot return as well, with a return trigger that may fire repeatedly and both parameters tuned jointly; together with the equal-data three-action dynamic program DP_N^3 , it separates the value of *pricing* two levers from the value of *having* them.

Results (Table 2, Table 5). For the handoff decision alone the reviewer’s expectation is borne out: the position-dependent threshold comes within 0.1–3.3 points of BATON-ho on every gate (26.0% against 26.1% on DET), so a well-started search recovers most of the handoff boundary. The cost-scaled rollout behaves similarly. The two-lever rule captures part of the depot-return value but stays 1.4–10.3 points below BATON across the gates, and DP_N^3 ties BATON on the deterministic plans but collapses on the conservative ones, where breaches are rare and bins cannot pool information along the load axis. On the five conservative gates BATON’s advantage over the strongest of all these competitors is 6.5–10.3 points with intervals above zero (Table 3). We now also say explicitly, after Proposition 2, that the critique applies to the myopic break-even rule, and that a stop-dependent threshold can represent the optimal boundary but must then be searched.

Changes. Three new competitors and DP_N^3 in Section 4.1 and Table 1; new columns in Table 2, Table 5 and Table 7; discussion in the first and third observations of Section 4.2 and after Proposition 2.

Computational fairness. The methods use different amounts of training data and computational resources. The authors should report performance under comparable data and computational budgets, clearly separating offline training time from online decision time. The training, validation, and testing protocol should also be fully documented.

Response. We have (i) documented the protocol, (ii) made the data budgets explicit and varied them, and (iii) measured offline and online time separately. (i) The new “Protocol” paragraph in Section 4.1 states that every data-driven policy—BATON, the thresholds, the two-lever rule, the rule-based triggers, the rollouts, the restocking trigger, DP_N , DP_N^3 and the RL policy—sees exactly the same 10^3 training days per instance. Hyper-parameters of tuned competitors are selected on realised cost over those same days; there is no separate validation set, which if anything favours the tuned competitors, since BATON has no continuous hyper-parameter (its one discrete choice, the recourse menu, is made on the same days). The 2×10^3 test days are used only for evaluation, and the seeds are fixed functions of the instance name. The only policies with more data are the two high-data reference programs, which are yardsticks rather than competitors; every table that reports them labels them, and the clairvoyant bounds, as reference points, and the main tables separate them from the competitors. (ii) The new data-budget experiment (Section 4.7, Table A.4 and Figure 7) varies the history from 100 to 20,000 days per route for BATON, the tuned threshold and the equal-data dynamic program. BATON beats the tuned threshold at every history length, down to 100 days per route, and approaches (on the city routes, at 20,000 days, slightly exceeds) the binned high-data programs as the history grows; those programs are estimates, not exact optima, and we have relabelled them accordingly. This result also led us to withdraw the recommendation in the original Remark 1 to fall back to a threshold for histories below about 500 paths. (iii) Table A.5 reports single-core offline fitting

times for every policy, with tuning included (median 24 ms per route for BATON), and the online decision time, which is $1.2\ \mu\text{s}$ per decision for every cost-comparison policy. The RL policy is now retrained on the same CPU; its training takes 1–3 minutes for the 49 routes, against 1.5 seconds for fitting BATON on the same routes.

Changes. “Protocol” paragraph in Section 4.1; new Section 4.7 with Table A.4, Table A.5 and Figure 7.

Statistical analysis. The experimental unit and dependence structure should be clarified. Paired confidence intervals, effect sizes, and route-level results would be more informative than extremely small p-values alone. Since the problem concerns costly emergencies, tail-risk measures would also be useful.

Response. We agree, and the original pooling of 240 plan comparisons across gates was inappropriate: the six gates re-use the same forty instances. The revised analysis (“Statistical analysis” in Section 4.1) takes the instance within a gate as the experimental unit. Routes of a plan share their test days and are aggregated within the plan, and each gate is analysed separately ($n = 40$). Table 3 reports, per gate and against the strongest competitor, BATON-ho and the high-data three-action program: the mean paired difference in saving with a 95% bootstrap confidence interval, the number of instances won, the matched-pairs rank-biserial effect size, and a Holm-adjusted Wilcoxon test. Because the strongest competitor is chosen on the same instances, its interval re-selects the competitor in every bootstrap resample, and its p -value is the largest of the p -values against each candidate competitor (an intersection-union test), so the selection does not inflate the evidence; the same selection-aware interval is used in the robustness tables. Against the strongest competitor the advantage is 6.5–10.3 percentage points on the five conservative gates, every interval excluding zero; on the deterministic gate BATON and the equal-data three-action program are tied ($\Delta = +0.0, [-0.4, +0.4]$). The key contrast of each robustness table (Table 5, Table 7, Table 10, and the day-type comparison in the text) now carries a paired bootstrap interval as well, and the text reports the narrow ties and the one loss with their intervals. Route-level results for every policy, gate and instance are released as supplementary data. For tail risk, Table 4 reports the CVaR₉₅ of the daily plan-level bill and the daily probability that any route of a plan suffers an emergency; BATON reduces the CVaR₉₅ by 22.3%–54.3% relative to the reactive policy.

Changes. “Statistical analysis” paragraph in Section 4.1; new Table 3 and Table 4; the pooled p -values have been removed, and the RL comparison is now tested at the instance level; route-level CSV files in the supplementary material and the public repository.

Methodological positioning. The manuscript should distinguish more clearly between the novel modeling contribution, the structural results, and the adaptation of established regression-based optimal-stopping methods. The novelty claims should be stated accordingly.

Response. We have rewritten the contribution statement in the introduction as three separate contributions of different kinds. The first is a *model*: execution-stage recourse in simultaneous pickup-and-delivery as optimal stopping over a menu of levers priced per stop from route geometry. The second is a set of *structural results* (Propositions 1–4). The third is a *method* that we explicitly describe as an adaptation of regression-based backward induction (Longstaff and Schwartz; Tsitsiklis and Van Roy), for which we claim novelty only for the pieces the model requires: the isotonic step licensed by Proposition 4, the simulation-based valuation of the post-return state, and the train-set selection of the recourse menu. We now also cite the classical antecedents of the structural results: monotone optimal policies for Markov decision processes

(Serfozo, 1976; Puterman, 1994) and stochastic-order comparison methods (Müller and Stoyan, 2002), of which Proposition 4 is an instance; the per-stop threshold structure of optimal restocking on a fixed route (Yang et al., 2000); and the fixed-sequence dynamic programs for pickup and delivery with depot returns of Minis and Tatarakis (2011) and Tatarakis and Minis (2009). We corrected the convergence claim: the analysis of Clément et al. covers a fixed linear basis, and the manuscript now points to the nonparametric analyses of Egloff (2005) and Zanger (2013) without claiming a convergence theorem for BATON, noting that the regression target is monotone only in the limit because it is computed under the fitted downstream policy. We have also toned down several formulations (for example “the most comprehensive computational study”, “significance levels that leave no room for doubt” and “no tuning”).

Changes. Introduction, paragraphs 5–7; abstract; related work (recourse strand); remark after the proof of Proposition 4; Section 3.4.

Scope of the conclusions. Some managerial conclusions should be qualified because the model does not include time windows, heterogeneous delay penalties, shared-resource constraints, or integrated route planning.

Response. Agreed. The conclusion now contains a paragraph “Scope of the conclusions” that names exactly these limits. The model has no time windows and prices lateness uniformly per downstream customer. Routes are fixed and executed in the planned order, and execution is not integrated with planning. Shared resources enter only through the standby pool of Section 4.6, and demand and prices are calibrated rather than taken from one operator. The paragraph also explains why we believe the *comparisons* between policies are robust to these choices, while the absolute savings are indicative. The managerial findings themselves are now qualified where needed. For example, the threshold critique is stated for the myopic break-even threshold, with the conditions under which a threshold is good enough, and the value of the depot return is attributed to plan slack and the demand profile. We had also attributed its low value on the city instances to urban geography; re-planning the city instances under the SAA gate to test this produced plans on which breaches all but vanish (reactive cost 0.04 per plan-day), so the two explanations cannot be separated with our instances, and the manuscript no longer claims the geographic one.

Changes. Conclusion, new paragraph “Scope of the conclusions” and revised managerial findings.

Reviewer 2

Independence assumption vs. experimental design. Assumption 1 (independent increments) underpins the Markov sufficiency of the scalar load and the monotonicity proof in Proposition 3. Yet the experiments use a Gaussian copula with equicorrelation $\rho = 0.6$. The authors should clarify whether the backward induction and Proposition 3 remain valid under dependent increments, or whether the method proceeds as if independence holds. A sensitivity analysis varying ρ (e.g., 0, 0.3, 0.6, 0.9) would substantially strengthen the robustness claim. More directly: if Proposition 3 requires independence, is isotonic regression still justified in the $\rho = 0.6$ experiments?

Response. Thank you for pressing on this point. Our answer has three parts. First, the proposition (now Proposition 4) does not require independence. Its proof needs only that the load be a Markov chain with stochastically monotone kernels (revised Assumption 1). This covers independent increments and the Gaussian day-factor model, in which W_k is sufficient for the

factor. Second, the benchmark copula model satisfies this only approximately, because pickups and deliveries load on separate factors with customer-specific scales. In that case the method proceeds as regression-based stopping in the (k, W_k) state, and the isotonic step is the monotone projection of $\mathbb{E}[\cdot \mid W_k]$. We now say this explicitly rather than leaving it implicit. Third, we ran exactly the sensitivity analysis you suggest, $\rho \in \{0, 0.3, 0.6, 0.9\}$, plus a multiplicative day factor (Table 7), and tested the shape itself on fifty thousand paths per route. The significant-violation rate of monotonicity is 0.2% under independence (the false-positive level of the test), 0.1% at $\rho = 0.3$, 0.1% at $\rho = 0.6$, 0.0% at $\rho = 0.9$ and 0.6% under the day factor. To show that the test is not blind, we injected a dip of 10% and of 25% of the mean continuation cost into one bin; it is detected in only 64% and 89% of the tests. The test therefore finds no significant violations, but with limited power: small dips could go unnoticed. We read the result as consistent with a monotone continuation value, and the isotonic restriction as a reasonable approximation where the theorem does not strictly apply, rather than as a proof of monotonicity. A BATON variant keeps the highest saving among implementable policies under all five laws. BATON itself is tied with its strongest competitor in three of the ten law–gate rows and loses one, at $\rho = 0.9$ on DET plans, to the equal-data three-action program (-1.1 points, [-1.6, -0.7]); the state-conditional variant reverses that loss. On the DET plans BATON’s share of the high-data three-action saving is 100%, 94% and 93% at $\rho = 0, 0.6$ and 0.9 , and under independence it attains 94% of the *exact* optimum, which we now compute by numerical convolution (Table 8). Interestingly, all implementable policies save *more* as ρ grows: correlation makes the running load informative about the rest of the day.

Changes. Revised Assumption 1 and discussion (Section 3.2); proof of Proposition 4 and the paragraph after it (Section 3.4); Section 4.5 with Table 7 and Table 8.

Bias in the fresh-start value estimator. In §3.5, $F_k = C_k(0)$ is estimated by simulating the reset state forward through the already-fitted downstream policy, not the optimal policy. Since the fitted policy is weakly suboptimal, the estimator is an upper bound on the true F_k , which systematically overprices $R_k + F_k$ and biases the policy away from depot return—potentially toward handoff or continuation. The authors should discuss the direction and magnitude of this bias, and quantify it in settings where depot return should be valuable (e.g., Salhi–Nagy CMT geometry).

Response. This is an excellent question, and answering it led to an improvement of the method. The reviewer is right that the suboptimality of the downstream policy pushes $\hat{F}_k$ upward. In the revision we identify two further sources of bias, of opposite sign, and quantify all three against a high-data reference fresh-start value computed from fifty thousand independent paths (Section 3.5 and Table A.3). (i) The suboptimality you describe pushes $\hat{F}_k$ up. We note that this is also the value the deployed policy will actually realise after a return, so it is the consistent price for the policy that is deployed. (ii) The forward pass runs on the training paths on which the downstream models were fitted, which pushes $\hat{F}_k$ down by in-sample optimism. (iii) Under positively dependent demand, returns are chosen on heavy days whose remainder is also heavy, so an unconditional F_k under-prices the return on exactly the days on which it is used. On Salhi–Nagy, the geometry you suggest, the in-sample error of $\hat{F}_k$ against the reference value is +0.50% of the handoff price and the out-of-sample error of the same fitted policy is +0.09%. Replacing $\hat{F}_k$ by the reference value changes the three-action saving from 52.2% to 51.9%. Sources (i) and (ii) are thus small and nearly cancel. Source (iii) is the material one, and it appears on delivery-dominated city routes, where returns rarely pay. It has a direct remedy within the method: regressing each training path’s realised post-return cost on its own pre-return state gives a state-conditional fresh-start value $\hat{F}_k(W_k)$, monotone for the same reason as C_k and flat under independence. The resulting BATON-CF needs no deployment selection and improves on

BATON on all six Dethloff gates (Table 2) and in 8 of the ten rows of Table 7. Since it was introduced during the revision, after the city results had revealed bias (iii), we report it beside BATON in the main tables, say so in the text, and keep BATON as the headline policy so that the main tables remain comparable with the submitted version. The high-data three-action reference DP_{50k}^3 uses the same state-conditional fresh-start value, which the manuscript now documents.

Changes. New paragraphs “Estimating the fresh-start value” and “Deployment selection” in Section 3.5; Section 4.5 and Table A.3; BATON-CF in Table 2, Table 5, Table 7 and Table A.5.

Action set and the value of the third lever. The paper only considers two levers: full handoff of the remaining route and depot return. Partial handoff—transferring only the riskiest remaining customers—is mentioned as future work but is likely common in practice. The claim that the action set is “general in form” needs clearer boundaries. Moreover, Table 4 shows that in real city instances BATON performs essentially the same as its handoff-only restriction (10.8% vs. 10.7%), implying that depot return is rarely useful on real road networks. The authors should discuss this more explicitly: the three-action policy’s value appears mainly in synthetic benchmarks and conservative planning gates, not in real urban settings.

Response. We agree on both counts. On the boundaries: Proposition 4 and the algorithm cover exactly the actions that either end the route at a price known at the current stop, or reset the load to a known level at a known price. The handoff and the depot return are of these kinds; a partial handoff is not, because it changes the remaining route itself. The conclusion now states this. It also explains what a partial handoff would require: a state augmented by the split point, a price schedule for every split, and one regression per stop and split, at a cost quadratic rather than linear in the route length. On the value of the third lever: we now say plainly, in Section 4.3 and in the conclusion, that the depot return is valuable where plans carry slack and depots are central—the conservative Dethloff gates and the Salhi–Nagy geometry, where BATON saves 53.0% against 42.3% for BATON-ho—and that on our city instances it rarely pays. There BATON keeps the return in the menu on only 37% of routes and equals its handoff-only form (10.6% against 10.6%). We had attributed this to urban geography. To test that explanation we re-planned the city instances under the SAA gate, which leaves slack; but on those plans breaches all but vanish (reactive cost 0.04 per plan-day), so no policy has anything to save. The city results are equally consistent with the demand profile (delivery-dominated routes whose load drifts downward) and the lack of slack in DET plans—on the Dethloff DET plans the return also adds only 1.8 points. The manuscript now states the finding in these terms and no longer claims that urban geography as such makes the return worthless.

Changes. Conclusion (managerial finding four; “Limitations and next steps”); Section 4.3.

Quantitative characterization of Proposition 2. The proof shows that a fixed threshold over-triggers, but does not quantify by how much. The authors should discuss when the threshold family remains “good enough” and whether the cost of over-triggering can be small in practice. This would help readers judge whether the structural critique is mainly qualitative or also economically significant.

Response. We thank the reviewer for this suggestion, which led to a new result. The new Proposition 3 shows that the myopic stopping time never exceeds the optimal one, pathwise. The excess expected cost of the myopic rule equals $\mathbb{E}[(H_{\sigma^0} - C_{\sigma^0}(W_{\sigma^0}))\mathbf{1}\{\sigma^0 < \sigma^*\}]$, and is bounded by $\mathbb{E}[(H_{\sigma^0} - L_{\sigma^0})\mathbf{1}\{\sigma^0 < \sigma^*\}]$, where L is the clairvoyant cost of the rest of the day. Under flat prices this is at most the handoff price times the probability of an intervention on a day that

would have completed cleanly. The threshold is therefore good enough exactly when the states in which it fires almost surely lead to a breach and prices decline slowly—tightly packed plans and short routes—and poorer on plans that carry slack. We also corrected the scope of the critique: the proposition concerns the myopic break-even threshold, and the abstract and introduction no longer claim that every fixed threshold over-triggers. Table 6 evaluates both sides on our routes. The pathwise ordering holds on 100.0% of test days and the bound on 734/735 routes; on the remaining route the regret exceeds the bound by 0.06% of reactive cost, within Monte Carlo error. Pooled over routes, the myopic regret is 3.8% of reactive cost against a bound of 7.7% (the average of the route-level ratios is 5.1%), and the clean-day term is 90% of the bound, so unnecessary interventions are indeed what is paid for. The regret is smallest on the city plans (2.0%) and on short routes (2.7% for at most 12 stops), 3.8% on Dethloff DET plans, and larger on conservative SAA plans (6.8%) and longer routes. A threshold tuned on realised cost is a different object—tuning moves it away from the break-even level, so the proposition does not bound it—but its excess over BATON-ho, fitted on the same thousand days, is small on tight plans too: 1.0% of reactive cost on DET plans and 0.4% on the city plans, against 3.0% on SAA plans. The answer to the reviewer’s question is therefore: for the handoff decision alone, the cost of over-triggering is economically modest on tight plans—there a tuned threshold is good enough—and grows on plans with slack. The large differences in Table 2 on the conservative gates come mainly from the depot return, which a handoff threshold cannot price, and a two-lever threshold rule that is given the return recovers only part of them.

Changes. New Proposition 3 and discussion (Section 3.3); “The price of over-triggering” in Section 4.4 with Table 6; managerial finding two in the conclusion.

Planning–execution co-optimization. [...] Since this co-optimization is only an outlook, the authors should state whether they consider it the most important future direction and why it was not implemented here.

Response. Yes, we consider it the most important direction, because it attacks the breach risk that no execution policy can reach: the risk at the first stop of a packed route. We did not implement it for two reasons, which the conclusion now states. First, it changes the object of study. Plans would depend on the execution policy, so the design that isolates this paper’s contribution—comparing execution policies on common plans—would no longer be available. Second, it requires the fitted execution cost of every candidate route inside the planner’s inner loop, which is a separate computational question.

Changes. Conclusion, last paragraph.

Demand exchangeability. The model treats days as exchangeable and does not consider seasonality, day-of-week effects, or promotional days. [...] A brief discussion would be useful.

Response. We added both a discussion and an experiment (Table 9). One day in five is a promotion day with higher deliveries and considerably higher pickups, and we compare three strategies on the same history of 10^3 days: a pooled fit that ignores the day type; day-type-specific fits that split that history by type (about 803 normal and 197 promotion days), as an operator can when promotions are scheduled; and a stale fit that uses the normal days only and is applied unchanged on promotion days. On promotion days BATON saves 29.7% with the day-type-specific fit, 29.4% with the pooled fit and 28.3% with the stale fit. Over all days the day-type-specific fits are no better than the pooled fit (27.6% against 27.6%; per plan -0.5 points, [-0.8, -0.3]): what they gain on promotion days they lose on normal days by fitting on fewer paths.

The pooled policy copes because a promotion day reveals itself through a heavier running load, which the monotone continuation value prices whatever the day type. The conclusion draws the practical rule: keep known day types in one training history; fit separately only when each type has a long history of its own; and where the mix drifts, re-fit on a rolling window and monitor.

Changes. Section 4.5 with Table 9; conclusion, “Limitations and next steps”.

RL comparison. [...] The authors may briefly discuss whether other RL architectures (e.g., DQN, PPO, attention-based) might narrow the gap, without necessarily running additional experiments.

Response. We added this discussion to Section 4.4. Value-based methods such as DQN estimate the same continuation value with a generic function class, and we would expect them to approach, not exceed, the isotonic regression at equal data. Actor-critic methods such as PPO would likely close part of the gap by learning a critic that plays the role of $\hat{C}_k$. Attention- and graph-based architectures pay off when the state is a large combinatorial object, and would become relevant for partial handoffs or fleet-level coordination. We also state in the manuscript that our three seeds and two epoch budgets do not rule out that longer training, other hyper-parameters or another architecture would narrow the gap; sample efficiency is our interpretation for the architecture we tested. Finally, we retrained the RL baseline on a route bundle consistent with the current plans, on the same CPU as BATON, and report its training time.

Changes. Section 4.4, paragraph after Table A.2; related-work paragraph on learning.

Clairvoyant bound. BATON exceeds the handoff-only clairvoyant bound on four of six planning gates. Section 4.2 explains this correctly, but the explanation should appear more prominently in the abstract and conclusion to avoid confusing readers.

Response. Done. The abstract now says that, because the depot return enlarges the action set, BATON can exceed the clairvoyant bound of the handoff-only class. The explanation—the bound is clairvoyant only about the handoff lever, and a return can prevent a breach at a price no handoff can match—now appears in the introduction, in Section 3.6 where the bound is defined, in Section 4.2, and in the first paragraph of the conclusion. Every results table labels the oracle as handoff-only (“oracle-ho”). We have also added the clairvoyant bound for the full three-action menu (oracle³), which lies above every policy; BATON attains 52%–68% of its saving on the Dethloff gates.

Changes. Abstract; introduction; Section 3.6; Section 4.2; conclusion; table captions.

Zero-pickup stations. City instances use pickup fractions uniform on $[0.2, 0.8]$, excluding stations with zero pickup. Since deliver-only customers are common in practice, the authors should either justify this choice or add a sensitivity check with a nonzero fraction of zero-pickup stations.

Response. We added the sensitivity check. Every city instance now has two deliver-only twins in which the pickup of a random 25% or 50% of the customers is set to zero, and the twins are re-planned (Table 5). With fewer pickups the load rises less along the route, breaches are rarer, and all savings shrink: at 25% and 50% deliver-only customers BATON saves 4.9% and 3.1%, while the tuned threshold falls to -1.5% at 50%. At 25% BATON still leads its strongest competitor by +0.4 points ([+0.03, +0.61]); at 50%, where recourse is worth least, it is tied with the equal-data dynamic program (+0.0 points, [-6.9, +2.4]) and reaches only 47% of the high-data

reference—the lowest share in the study, which we report in the abstract’s range. Our original choice of strictly positive pickups followed the construction of the Dethloff benchmark, in which every customer has both a delivery and a pickup; the new twins show that the ranking survives deliver-only customers while the size of the prize, and BATON’s margin, shrink.

Changes. Section 4.1 (instances); Table 5; Section 4.3.

Figure 2. The three panels are small and may be difficult to read in print. Consider splitting into two figures or increasing font size.

Response. We did both: the explainer is now split into two figures (Figure 2, two panels at full text width, and Figure 3, the test-day costs), with all fonts enlarged to print size.

Changes. Figure 2 and Figure 3.

Reviewer 3

1. The paper divides the vehicles into three classes, the planned fleet, the standby class and the emergency class. What is the difference between the standby and the emergency vehicles? How many standby vehicles are there?

Response. The two classes differ in how they are contracted and in what they cost the customer, not in the vehicles themselves. A standby vehicle is reserved capacity under a pre-agreed rate: it is dispatched before a breach, takes over the remaining route at a small surcharge, and keeps the downstream customers on schedule. An emergency vehicle is hired on the spot market after the planned vehicle has overflowed: it pays surge rates, must first reach a stranded vehicle, and every downstream customer is served late. In the default model the standby pool is shared across the operation and billed per vehicle-day used, so the number of standby vehicles is not a constraint but a consequence of the policy. We now report it: the pool that covers 95% of days has a median of 3.5 and a maximum of 7 vehicles per plan for BATON on the Dethloff DET plans, and a median of 0 on the SAA plans, where the depot return does most of the work (over all six gates: median 0, maximum 7; 1 and 8 for the handoff-only restriction). In the new experiment with a pool of fixed size reserved in advance (Section 4.6), reserving vehicles pays only when they can be held for well under the standby day rate and handoffs are frequent: at a holding cost of 5 per vehicle-day the optimal pool is about 4.2 vehicles for a packed Dethloff DET plan and about 27.2 for a metropolitan pool serving some 66 routes, while slack plans need none.

Changes. New paragraph “Standby versus emergency vehicles” in Section 3.1; Section 4.6 and Table 10.

2. Since the paper does not consider the holding cost of the standby vehicles, if the number of the standby vehicles larger than the emergency vehicles, then the Assumption 2 and the Eq. 7 may not hold.

Response. Thank you; the original text did not make this clear. In the default model the holding cost enters through the standby day rate F_{sb} in the handoff price, charged per vehicle-day consumed. The revised Section 3.1 says so explicitly. When standby vehicles are instead reserved in advance, their holding cost is paid whether or not they are used; the new experiment of Section 4.6 models exactly this, with S reserved vehicles, a range of holding costs and the break-even holding cost of a reserved vehicle.

The reviewer is right that Eq. (7) ($E_k > H_k$) can fail if standby capacity is expensive. We have therefore checked what the theory actually needs, and it is less than we had assumed. The revised Assumption 2 requires only that the emergency schedule be non-increasing along the route; no ordering between H_k and E_k is used anywhere. The proof of Proposition 4 bounds the value function by the emergency price through the cost of never acting, not through H_k . Eq. (7) is now presented as a property of our calibration, not a requirement. When it fails, the handoff is simply dominated at the stops concerned and the policy never uses it there. To demonstrate this we added a configuration with $F_{sb} = 60$, in which the handoff price exceeds the emergency price at late stops (Table 11, the $F_{sb} = 60$ row). BATON saves 36.8% there, against 1.4% for the tuned threshold.

Changes. Revised Assumption 2 and the paragraph after it; text after Eq. (7); proof of Proposition 4; new row of Table 11.

3. In Sec 4.2, it is claimed that ‘BATON attains the lowest execution cost on every gate’, but in Table 2 and Fig. 4, it is indicated that in the case of ‘Deterministic’ and ‘W-DRO’, the value of ‘oracle’ is greater than that of ‘BATON’.

Response. Thank you; our wording was imprecise. The oracle is a clairvoyant policy: it knows the whole day’s demand in advance, so it cannot be implemented, and it serves only as a bound. The claim should have read “the lowest execution cost of every *implementable* policy”. It now does, everywhere (abstract, introduction, Section 4.2, conclusion), and the main tables now separate the reference points (the clairvoyant bounds and the high-data dynamic programs) from the competitors. The oracle is moreover restricted to the handoff lever, which is why BATON exceeds it on the other gates: a depot return can prevent a breach at a price no handoff can match. On the DET and WDRO gates this second lever is worth less, and the clairvoyant handoff-only policy stays ahead.

Changes. Section 4.2 (first and second observations); captions of Table 2 and Figure 5; abstract and conclusion.

4. In Table 3 and Table 5, only the value of ‘BATON-HO’ is displayed. What is the value of ‘BATON’?

Response. Both tables now show BATON as well. In the synthetic scenarios (Table A.1) the routes have no geometry from which to price a depot return; the submitted version therefore used the handoff lever only. We now price the return flat at half a handoff and report the full BATON, which reaches 69.1% on the collect-then-deliver scenario. In the RL comparison (Table A.2) the learned policy acts on the handoff lever only, so BATON-ho is the like-for-like comparison (32.9%); the full BATON, which adds the depot return, saves 33.7% on the same routes.

Changes. Table A.1 and Table A.2.

5. Table 4 shows that in the case of ‘City, real shops’ and ‘City, uniform’, the value of ‘BATON’ is smaller than that of ‘DP50k’ and ‘oracle’. Why?

Response. Thank you for pointing this out; the original tables did not make the role of these two columns clear. Both are reference points rather than competitors, and the revised tables separate them accordingly. The *oracle* knows the day’s demands in advance. On city routes much of the breach risk comes from demand surprises at the next stop that no policy observing only the past can anticipate, so its lead is large and unattainable. DP_{50k} is fitted on fifty times more data than

BATON. On the city instances the depot return rarely pays, so the comparison is between two handoff-only policies, and the gap (10.6% against 12.1%) is a gap to a reference that has fifty times more data and also a different estimator (binned means instead of isotonic regression), so it cannot be attributed to the data alone. The new data-budget experiment indicates that the amount of data accounts for most of it (Table A.4, Figure 7): given 20,000 training days, BATON saves 12.9% on the city instances, slightly more than DP_{50k} (12.2%; the binned program is an estimate, not an exact optimum). The oracle’s lead, by contrast, does not shrink with data: it is the value of knowing the future.

Changes. Section 4.3; Section 4.7 with Table A.4 and Figure 7; table captions.

Corrections found by the authors

1. **Proposition 1 (endpoint-label bias).** The proof of the cost bound stated that the optimal policy is “no worse than the clairvoyant policy”; the inequality runs the other way, so the claimed upper bound on the optimal cost and the “factor C_{fail}/ω_F ” were not proved. The corrected Proposition 1 bounds the loss of the endpoint-trained policy from both sides, with the upper bound attained when the breach is announced one stop ahead.
2. **Proposition 2 (over-triggering).** The strictness condition in the original statement was not sufficient: the policy used in the proof also hands off on days that would not have breached. The corrected Proposition 2 gives a condition that is proved—the path must reach, with positive probability, a later state in which the myopic rule itself fires. It also derives strict inclusion of the stopping regions under continuity, and it is now stated for position-dependent prices.
3. **A side condition in Proposition 1.** The formal verification showed that the lower bound on the loss needs routes with at least two stops (so that a decision epoch precedes the last stop) and $0 \leq \omega_F \leq C_{\text{fail}}$; the statement now says so.
4. **The “92–99% of a near-exact dynamic program” claim** in the original abstract did not hold on every gate: it referred to the handoff-only program, and on the WDRO and M-DRO gates the ratio was lower. Moreover the binned programs are estimates, not exact optima. The revised abstract compares BATON with the high-data *three-action* program on the Dethloff gates (86%–96%), with the exact optimum where it can be computed (94% on deterministic plans, 59% on slack ones), and gives the full range over all benchmarks and demand laws with a non-negligible reactive cost (47%–100%).
5. **“Every fixed threshold over-triggers.”** The original abstract, introduction and conclusion attributed to every fixed threshold what Proposition 2 proves for the myopic break-even rule. The claims are now stated for that rule; a threshold tuned on realised cost is not covered.
6. **Convergence citation.** The analysis of Clément et al. (2002), which we cited for the convergence of the backward induction, covers a fixed linear basis and not isotonic regression; the text now cites the nonparametric analyses of Egloff (2005) and Zanger (2013) and states that we do not prove a convergence theorem.
7. **Urban geography.** The original text attributed the low value of the depot return on the city instances to congested urban networks. Our instances cannot separate geography from plan slack and demand profile (see the reply to Reviewer 2, comment 3), and the claim has been withdrawn.
8. **Reproducible scenario seeds.** The evaluation derived its per-instance seeds from Python’s salted string hash, so scenario draws differed from run to run. Seeds are now fixed functions of the instance name, and all results were regenerated. Numbers moved by sampling noise (for

example, the tuned threshold’s saving on WDRO plans is now -1.3% rather than -8.3%); no qualitative conclusion changed.

9. **Cost-sensitivity baseline.** The baseline row of the sensitivity table was computed on all 40 instances while the sweep used 12; it is now computed on the same 12.
10. **RL baseline.** The route bundle of the RL experiment had been built from an earlier version of the city plans. It is rebuilt from the current plans (49 routes), and the RL policy is retrained on it. The description of the training now matches the code; the original text mentioned entropy regularisation and a learning-rate search, which the code did not perform.
11. **Timing.** The original text said “single-digit milliseconds” for fitting; the measured median for the full policy with deployment selection is 24 ms per route, which we now report.
12. **W-DRO gate.** With a type-1 Wasserstein ball and a 1-Lipschitz loss, the worst-case CVaR equals the empirical CVaR plus $\varepsilon/(1 - \alpha)$, so the WDRO gate is the SAA gate at a tighter threshold. We now say so in Section 4.1 and present it as a more conservative regime rather than an independent paradigm.

We hope that the revised manuscript meets the reviewers’ expectations, and we thank them again for their time and insight.

Sincerely,

Quang-Vinh Dang (corresponding author), on behalf of all authors